

Round 2 Technical Report

Data Vortex 2026 · Rebuilding the Social Engine's Semantic Layer

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Sentiment: logreg_union -- acc 0.6133, macro-F1 0.6134

Topic: svc_union -- acc 0.9677, macro-F1 0.8051

Every number in this report is computed by the Round 2 pipeline from Dataset 2 and the trained models -- nothing is typed by hand. Re-run `./run_round2.sh` to reproduce this PDF byte-for-byte (modulo timestamps).

1 · Problem definition

Round 1 restored the Social Engine's structured analytics; the semantic layer -- understanding tone and intent -- is still down. Dataset 2 provides 7,900 labelled social posts (text_id, post_text, sentiment_label, topic_category). We rebuild two supervised classifiers:

- Primary -- sentiment classification into Negative, Neutral, Positive.
- Secondary -- topic classification into 4 categories: Account_Security, Community_Discussion, Feature_Feedback, Technical_Issues.
- Bonus (our addition) -- unsupervised topic discovery (LDA/NMF) checked against the human topic labels, plus calibration, learning-curve and slice analyses.

Success criterion, fixed before training: macro-F1 on a held-out 20% test set -- macro (not accuracy) because the minority classes are the ones a semantic layer must not ignore. The test set is touched exactly once per task, by the refit CV winner.

2 · Preprocessing pipeline

One normaliser serves every model and ships inside the saved bundles, so inference can never diverge from training. Stages:

#	stage	what it does	why
1	export repair	decode literal \uXXXX escapes; unescape HTML entities (2x); fold curly quotes	the file carries its own corruption -- fixed before any linguistics
2	masking	URLs -> <url>, @mentions -> <user>, #tags -> words + <hashtag>	learn that a link existed, not 3,000 rare URL tokens
3	affect markers	emoticons -> <smile>/<sad>; emoji -> <emoji>; elongation -> doubled + <elong>	affect signals survive as features instead of punctuation noise
4	contractions	can't -> can not, 're -> are, ...	negation must survive stopword removal
5	stopwords	compact list, negations explicitly kept	noise out, sentiment in
6	negation scope	prefix next 3 content words after not/no/never/... with NOT_	not good must not vote positive (Pang & Lee style)
7	features	word TF-IDF (1,2-grams, min_df=2, sublinear) +/- char_wb TF-IDF (3-5)	words carry sentiment; characters carry typo/elongation robustness

Live demonstration -- real train posts through the shipped normaliser (raw -> normalised -> first tokens):

case	raw	normalised
URLs / mentions / hashtags	Conor McGregor, Grade-A fighter & Grade-A s**t talker. On December 12th, he's gonna take Aldo's belt. #TheNotorious	conor mcgregor grade a fighter grade a s t talker on december 12th he gonna take aldo belt thenotorious <hashtag>
negation	Man, I am going to change my vote based on Axl Rose's opinion said no one ever. Politico: C'MON MAN	man i am going to change my vote based on axl rose opinion said no one ever politico c mon man
emoji / elongation	"\''''''@PunkinPwincess: David Guetta is coming to Fort Canning at 11 October :3 OMG OMG OMG! :3\'''''' Will b going :D"	<user> david guetta is coming to fort canning at 11 october 3 omg omg omg 3 will b going <smile>
ordinary post	Conor McGregor, Grade-A fighter & Grade-A s**t talker. On December 12th, he's gonna take Aldo's belt. #TheNotorious	conor mcgregor grade a fighter grade a s t talker on december 12th he gonna take aldo belt thenotorious <hashtag>

case	word tokens (first 12)
URLs / mentions / hashtags	conor mcgregor grade fighter grade s t talker december 12th gonna take ...
negation	man going change vote based axl rose opinion said no NOT_ever NOT_politico ...
emoji / elongation	<user> david guetta coming fort canning 11 october 3 omg omg omg ...
ordinary post	conor mcgregor grade fighter grade s t talker december 12th gonna take ...

3 · Model selection

Six specs form a ladder: chance floor (stratified dummy), classic baseline (MultinomialNB on words), linear workhorses (LogReg / LinearSVC on words), then the same two on words+characters. The word-vs-union gap is the ablation that prices the character arm. Why linear-only: 9k short texts do not justify a GPU model; linear TF-IDF models are reproducible anywhere, inspectable, and the honest baseline this task asks for. Grids are small (C in [0.5, 1.0, 2.0, 4.0] / [0.25, 0.5, 1.0, 2.0], NB alpha in [0.1, 0.5, 1.0], class_weight in {None, balanced}) -- compared, not assumed.

CV results -- sentiment

model	CV macro-F1	best params
logreg_union ★ winner	0.6110 ± 0.0148	C=0.5, class_weight=balanced
svc_union	0.6028 ± 0.0091	C=0.25, class_weight=None
logreg_word	0.5915 ± 0.0156	C=1.0, class_weight=balanced
svc_word	0.5893 ± 0.0136	C=0.25, class_weight=balanced
nb_word	0.5709 ± 0.0120	alpha=0.5
dummy	0.3240 ± 0.0067	—

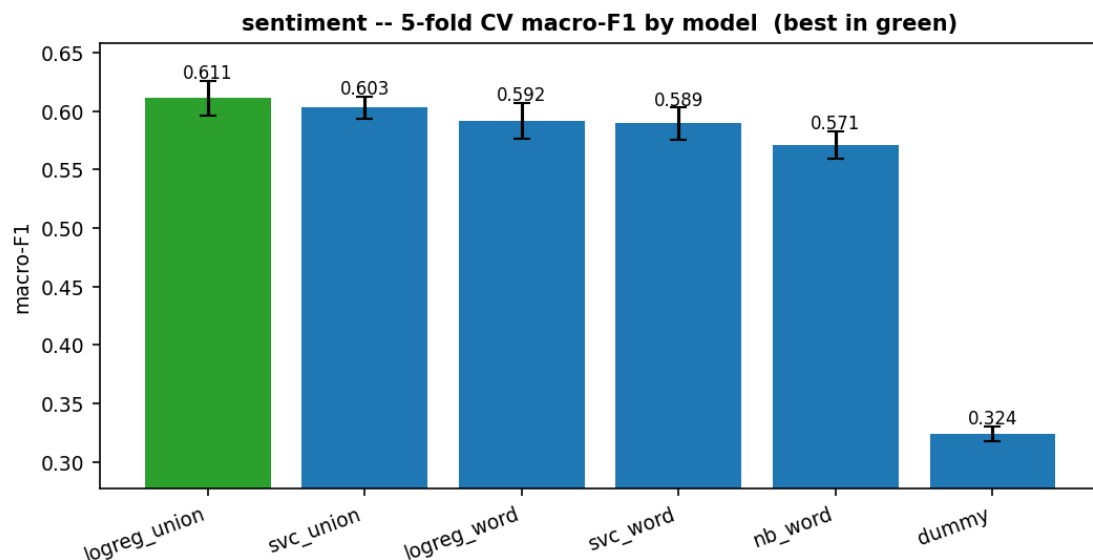


Figure: sentiment CV macro-F1 by model.

CV results -- topic

model	CV macro-F1	best params
svc_union ★ winner	0.7800 ± 0.0374	C=2.0, class_weight=balanced
logreg_union	0.7666 ± 0.0339	C=1.0, class_weight=balanced
svc_word	0.6042 ± 0.0361	C=2.0, class_weight=balanced
logreg_word	0.5791 ± 0.0278	C=4.0, class_weight=balanced
nb_word	0.3768 ± 0.0452	alpha=0.1
dummy	0.2397 ± 0.0112	—

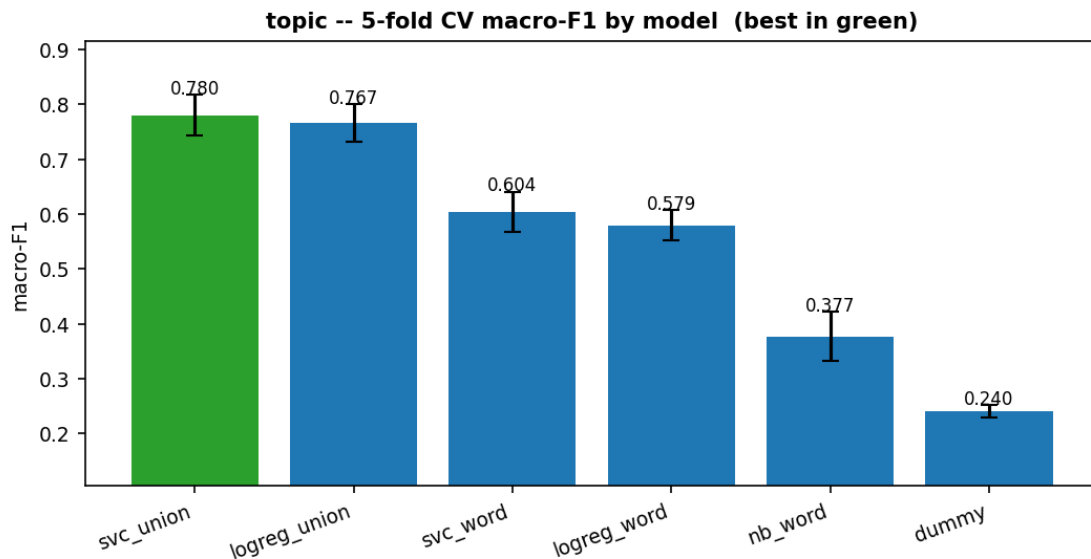


Figure: topic CV macro-F1 by model.

4 · Training methodology

- One shared stratified split (sentiment_x_topic, seed 42): 6,320 train / 1,580 test, id overlap 0. Both tasks share it, so their scores are comparable.
- Grid search with 5/5-fold stratified CV (sentiment/topic) on train only, scoring macro-F1; ties broken toward the simpler spec.
- Winner refit on full train (GridSearchCV refit), evaluated ONCE on test.
- Sentiment winner ships sigmoid-calibrated (3-fold, train only) because LinearSVC has no native probabilities; topic winner likewise.
- Unsupervised LDA/NMF (K=4) fit on train texts with no labels; agreement scored on test.
- Total train time 273.2s; one seed (42) drives splits, folds and samplers; environment recorded in the metrics report.

5 · Evaluation metrics

metric (test)	sentiment	topic
accuracy	0.6133	0.9677
macro-F1 ★ primary	0.6134	0.8051
weighted-F1	0.6117	0.9631
MCC	0.4201	0.8645
ROC-AUC OvR	0.7916	0.9789
log-loss	0.8621	0.1163
Brier OvR	0.1703	0.0140

Per-class -- sentiment (logreg_union)

class	precision	recall	F1	support
Negative	0.6337	0.6728	0.6527	486
Neutral	0.5655	0.5191	0.5413	549
Positive	0.6375	0.6550	0.6462	545

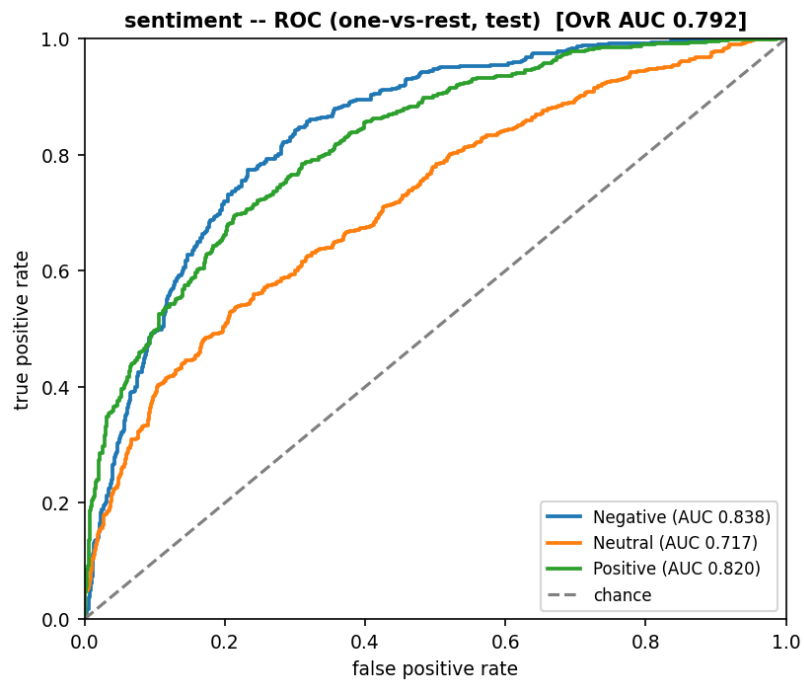


Figure: sentiment ROC, one-vs-rest (test).

Per-class -- topic (svc_union)

class	precision	recall	F1	support
Account_Security	1.0000	0.5000	0.6667	26
Community_Discussion	0.9652	0.9993	0.9819	1,360
Feature_Feedback	1.0000	0.4400	0.6111	50
Technical_Issues	0.9854	0.9375	0.9609	144

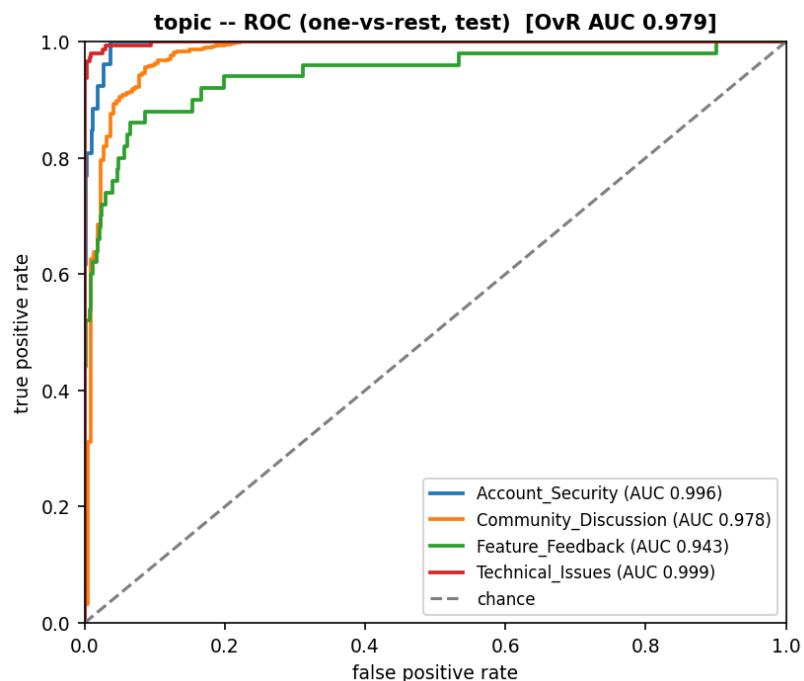


Figure: topic ROC, one-vs-rest (test).

6 · Confusion matrix

Counts show where the errors are; row-normalised (recall) shows which classes the model refuses to predict. Read with the per-class table above: a rare class with high precision but low recall is being under-predicted, not mispredicted.

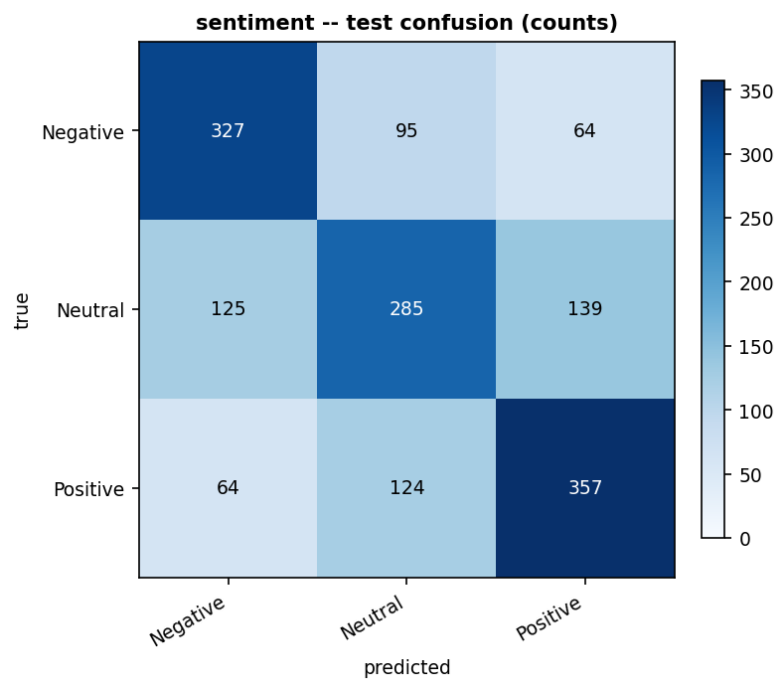


Figure: sentiment confusion (counts, test).

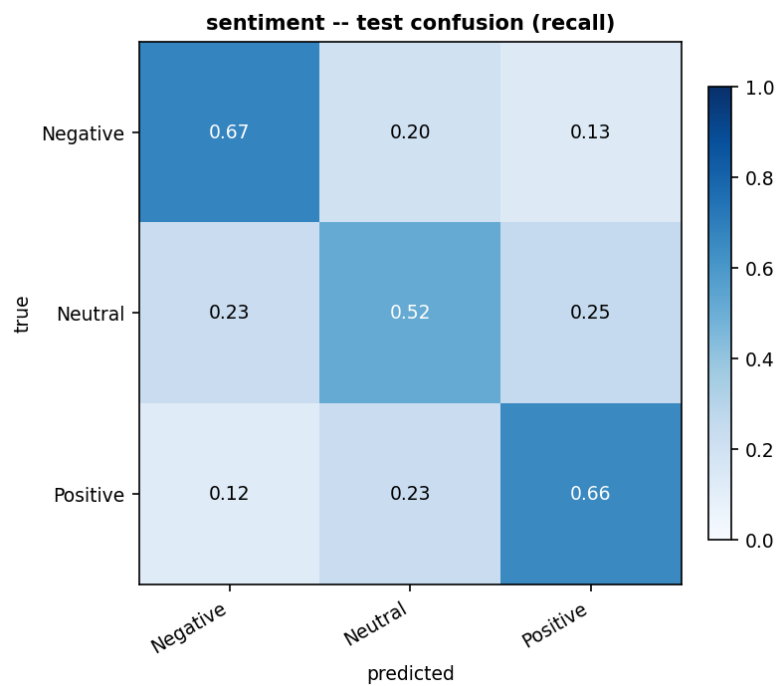


Figure: sentiment confusion (row-normalised = recall).

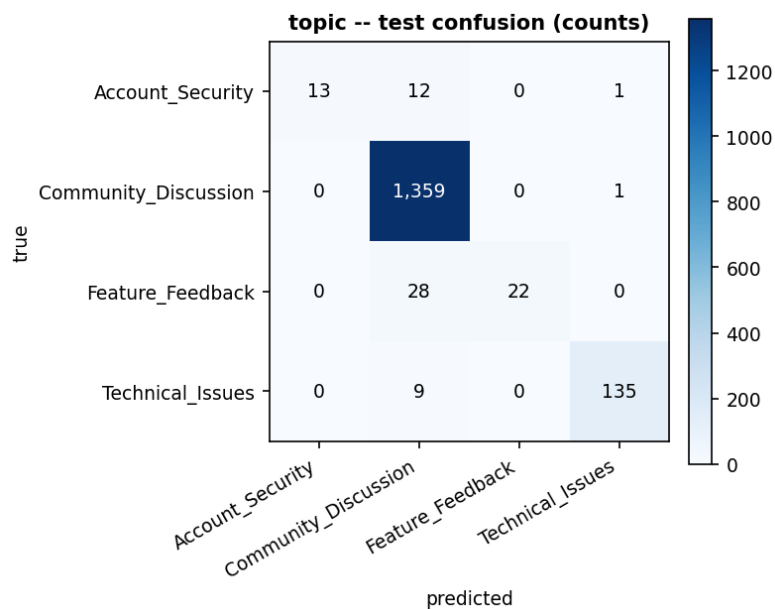


Figure: topic confusion (counts, test).

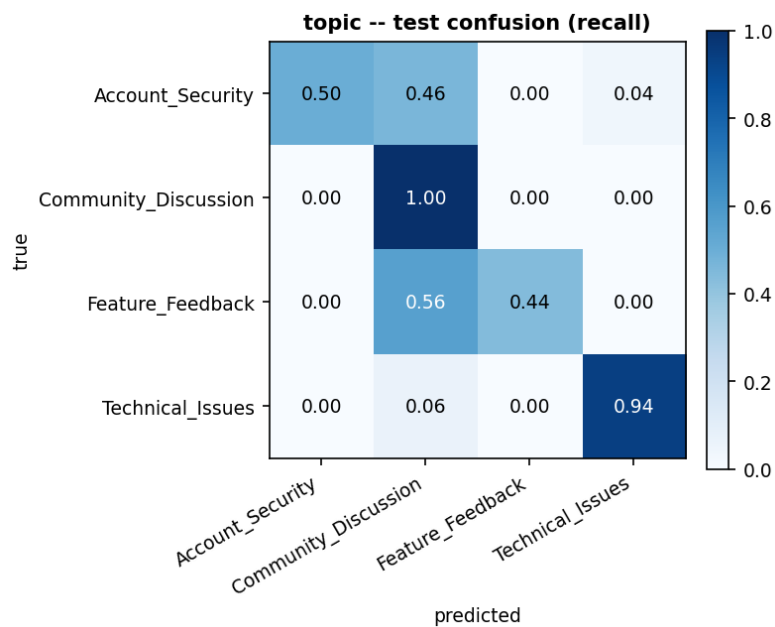


Figure: topic confusion (row-normalised = recall).

7 · Error analysis

7.1 · sentiment -- measured slices

- Negation present vs absent: negation_yes 0.608 (n=245) vs negation_no 0.614 (n=1,335) --> -0.006.
- Emoji/emoticon present vs absent: emoji_yes 0.754 (n=61) vs emoji_no 0.608 (n=1,519) --> +0.146.
- URL present vs absent: url_yes 0.500 (n=2) vs url_no 0.613 (n=1,578) --> -0.113.
- Length extremes: len_short 0.571 vs len_long 0.689 --> spread 0.118.

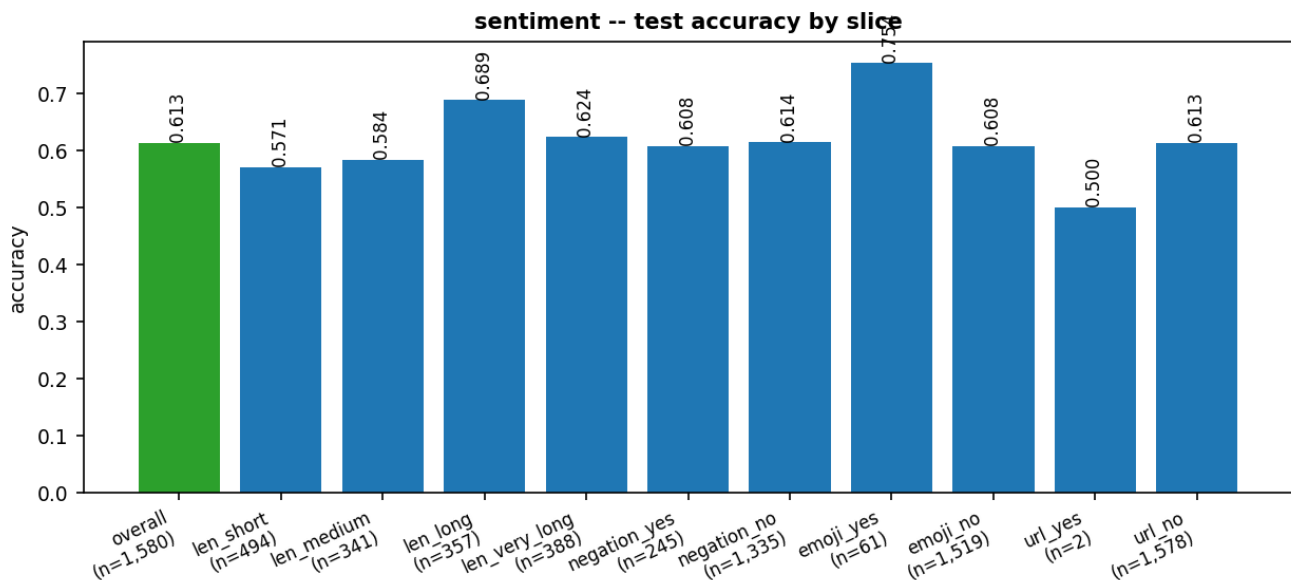


Figure: sentiment accuracy by slice (test).

sentiment -- real mistakes (top confused pairs)

verbatim cleaned posts the shipped model got wrong -- the qualitative half of the analysis. IDs let judges re-find each row in the test split.

true → pred	post text (cleaned)	id
Neutral → Positive	now watching Denzel Washington talking about his new movie flight it's comming out Nov 2nd 2012 #LateShow With David Letterman	TXT_00082
Neutral → Positive	I'm selling my 2 Ed Sheeran tickets for Gillette Stadium September 25th both tickets would come to \$200 total lmk if you're interested!	TXT_00182
Neutral → Negative	"If you wonder why we keep on this sustainable path, run the trails around Cardigan. Closed loop at its best, nature. #doworkshop tomorrow"	TXT_00276
Neutral → Negative	I never expect too much. Expect too much and you may be disappointed. This is referring to this FRANK OCEAN release. Just chill n' wait guys	TXT_00319
Positive → Neutral	9 months from Friday there is going to be another ginormous Baby Boom #CallinIt	TXT_00029
Positive → Neutral	The game 1st album 1 song he say he rap like Eazy-E then Nas then 50 cent then dr Dre then Jay-Z then Eazy-E again then like his dead bro	TXT_00033
Negative → Neutral	November 21 in the lonely hour tour.. OKAY SAM SMITH OKAY!!!! :-(((TAPOS SOLD OUT PA OKAY!!!! OKAY LANG TALAGA	TXT_00108
Negative → Neutral	@user u 2 missed it. statement was plotted to take attention away from UK national Rahul Gandhi. 2ndly, such BS ll hit fan in Apr."	TXT_00170

7.2 · topic -- measured slices

- Negation present vs absent: negation_yes 0.976 (n=245) vs negation_no 0.966 (n=1,335) --> +0.009.
- Emoji/emoticon present vs absent: emoji_yes 0.967 (n=61) vs emoji_no 0.968 (n=1,519) --> -0.001.
- URL present vs absent: url_yes 1.000 (n=2) vs url_no 0.968 (n=1,578) --> +0.032.
- Length extremes: len_very_long 0.946 vs len_short 0.984 --> spread 0.038.

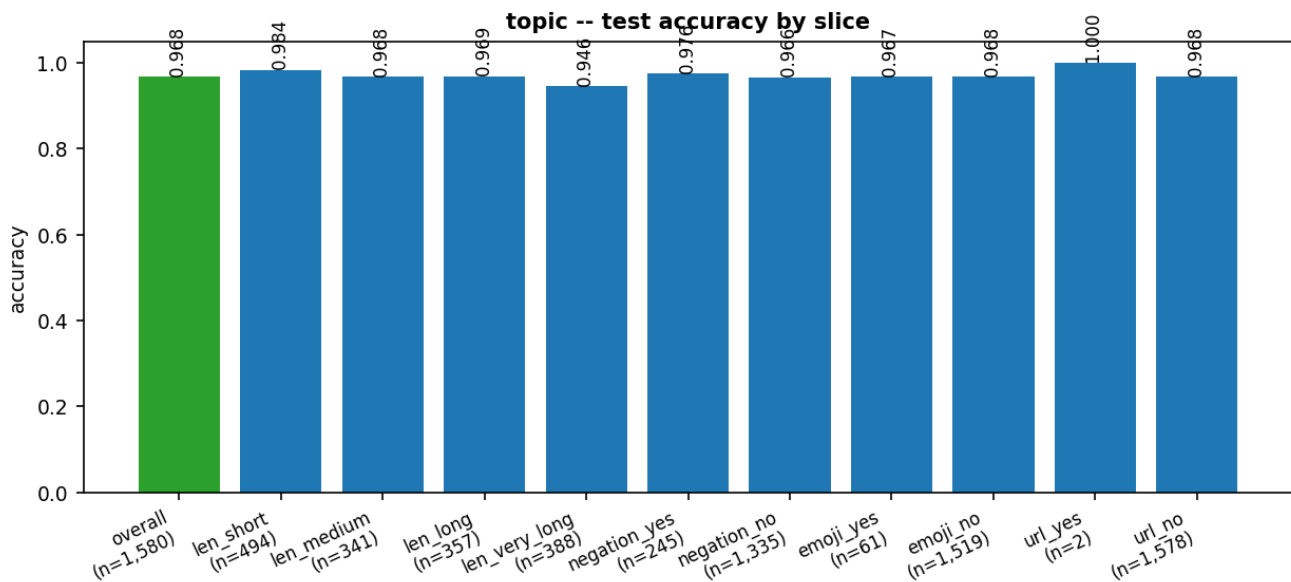


Figure: topic accuracy by slice (test).

topic -- real mistakes (top confused pairs)

verbatim cleaned posts the shipped model got wrong -- the qualitative half of the analysis. IDs let judges re-find each row in the test split.

true → pred	post text (cleaned)	id
Feature_Feedback → Community_Discussion	Sunanda Pushkar's son Shiv Pushkar Menon on Saturday asked the Delhi Police to conclude the inquiry into his mother's death expeditiously...	TXT_00492
Feature_Feedback → Community_Discussion	"fuck everything, i will go see Paper Towns tomorrow and nobody can ruin my day"	TXT_00761
Account_Security → Community_Discussion	@user @user reported in Oct: IMF has broken the good news that #IslamicState Saudi Arabia is going bankrupt within 5 years	TXT_01372
Account_Security → Community_Discussion	So excited for Lolla Friday because so many amazing bands will be there oh and someone named Paul McCartney??? Idk if he'll be good tho	TXT_01689
Technical_Issues → Community_Discussion	We're supposed to believe a pap recognised and papped Briana on Oct 26 then just sat on the pics til Nov 3 Imfao PLEASE	TXT_00458
Technical_Issues → Community_Discussion	"Left over Halloween candy is probably past it's expiration, right? That's what I'll tell my kids when they see the empty wrappers tomorrow."	TXT_00978
Community_Discussion → Technical_Issues	As of tomorrow #HTC & @user are a month late (or a year) in updating my Thunderbolt to #ICS. And phones with #jellybean are available.	TXT_03699

8 · Bonus analyses (our additions)

8.1 · Do discovered topics match human topics?

LDA and NMF discover topics with no labels; we check them against the human topic_category on test texts. NMI/ARI near 0 would mean the human taxonomy is invisible to word co-occurrence; high values validate it.

model	NMI	ARI	homogeneity	completeness
LDA	0.0020	-0.0030	0.0035	0.0014
NMF	0.0027	0.0068	0.0046	0.0019

Unsupervised-vs-human agreement (row-normalised, counts in brackets)

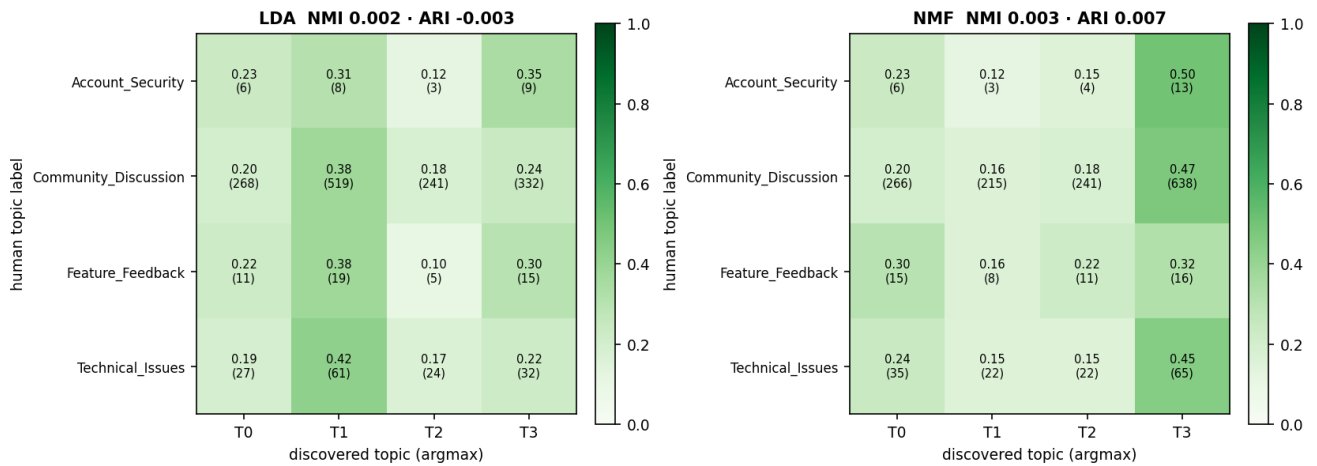


Figure: human vs discovered topic agreement.

Discovered topics -- LDA top-10 words (ranked)

LDA T0

1. <user>
2. <hashtag>
3. time
4. day
5. 3rd
6. 1st
7. sunday
8. mon
9. c
10. thursday

LDA T2

1. tomorrow
2. game
3. <hashtag>
4. <user>
5. 1st
6. sunday
7. 2
8. 1
9. 2nd
10. day

LDA T1

1. <user>
2. tomorrow
3. not
4. going
5. friday
6. see
7. go
8. day
9. night
10. make

LDA T3

1. <hashtag>
2. not
3. <user>
4. night
5. tomorrow
6. day
7. saturday
8. time
9. last
10. thursday

Figure: LDA top-10 words per discovered topic.

8.2 · Would more labels help?

Learning curves (3-fold CV macro-F1 vs train size) separate a data bottleneck (rising validation curve) from model saturation (plateau).

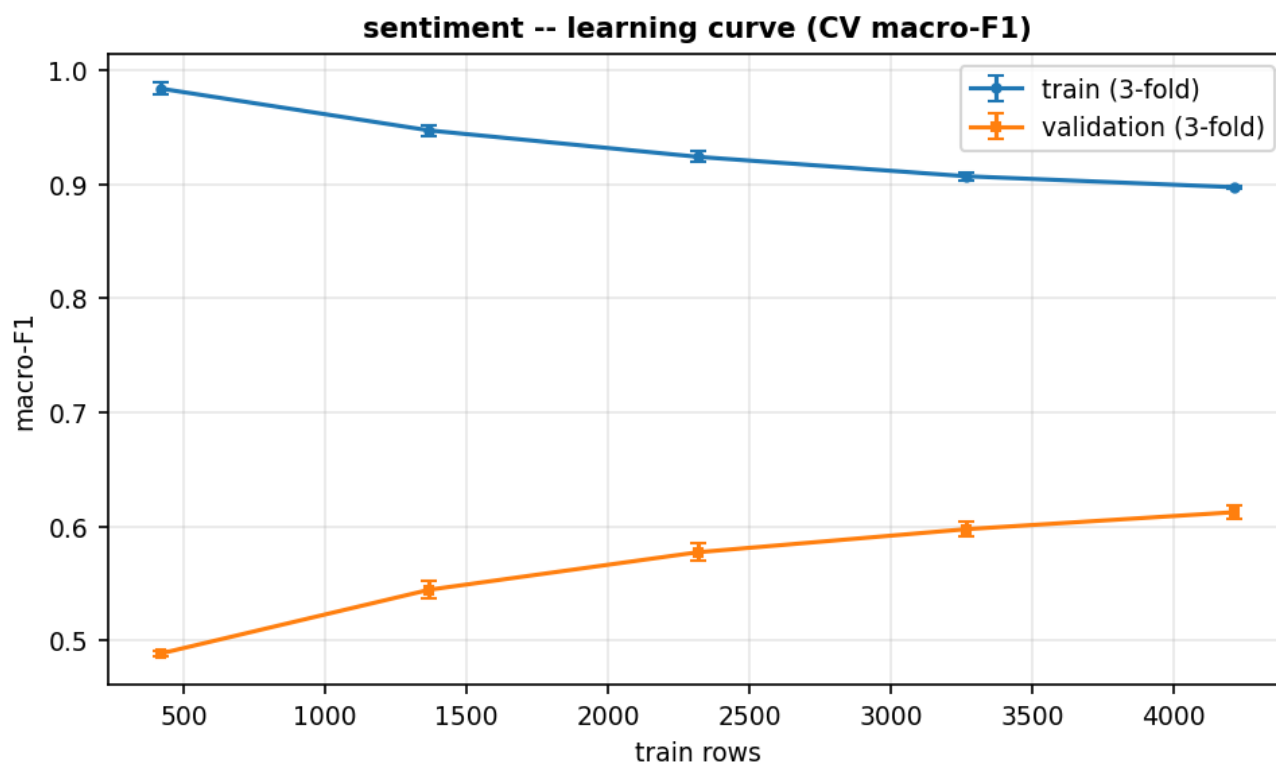


Figure: sentiment learning curve.

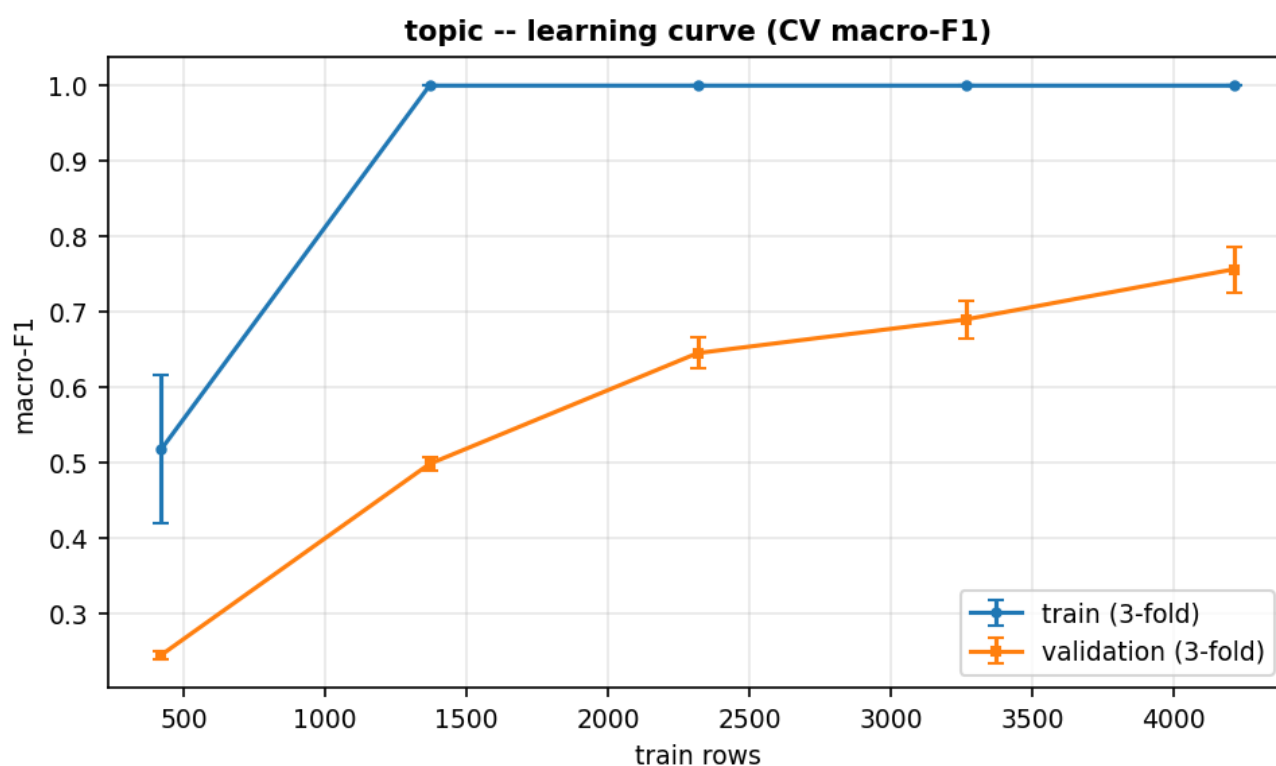


Figure: topic learning curve.

8.3 · Can we trust the probabilities?

Reliability diagrams plot predicted vs empirical probability per class; the Brier scores are in §5. Well-calibrated probabilities are what let the Social Engine threshold (act only when confident) instead of argmaxing blindly.

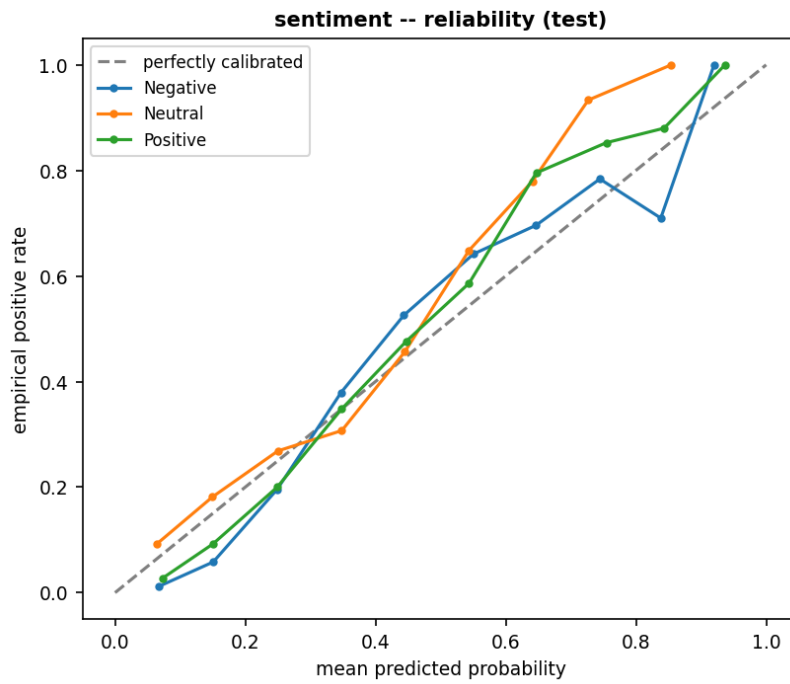


Figure: sentiment reliability (test).

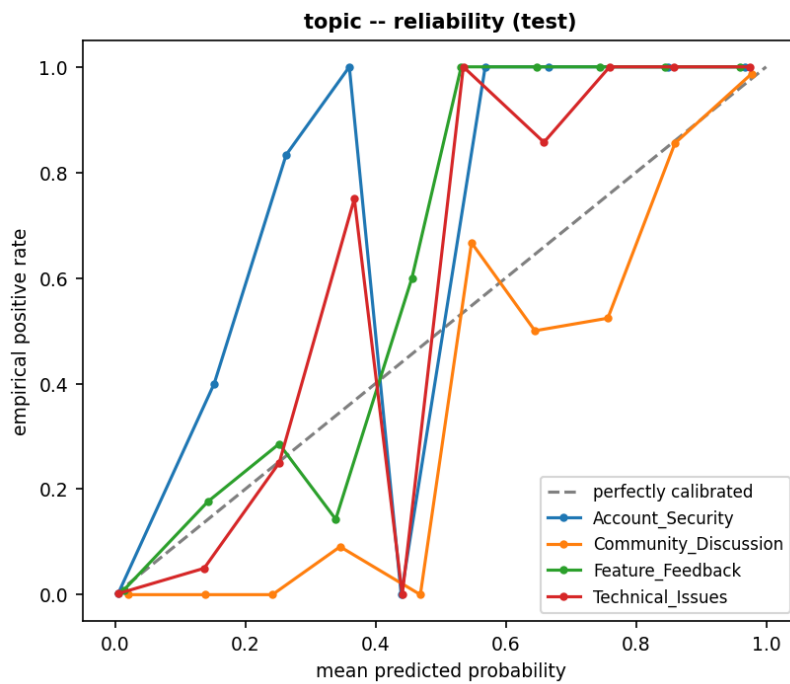


Figure: topic reliability (test).

8.4 · Surface entity census (regex baseline)

signal	posts	share
URL	14	0.2%
@mention	1,871	29.6%
#hashtag	1,131	17.9%
emoji/emoticon	228	3.6%
negation word	940	14.9%

Framed honestly: this is a pattern-based entity census over train, not a trained NER -- it quantifies what a future NER layer would have to recognise (handles, tags, links). A supervised NER head is listed under future work, not claimed as done.

9 · Limitations & threats to validity

- Label noise: 0 distinct texts appear with conflicting labels -- an upper bound on achievable accuracy no model can cross.
- Short-text ambiguity: one-liners with no context (names, scores, event mentions) are genuinely hard; the length slices in §7 quantify it.
- No external pretraining by design: linear TF-IDF cannot resolve sarcasm or world knowledge; misclassified examples in §7 show the boundary.
- Domain shift: trained on this export's style (escapes, entities); new corruption modes would need the normaliser extended, not the models retuned.
- Topics are imbalanced (see §5 support column); macro-F1 keeps the rare topics honest, but their confidence intervals are wide.

10 · Reproducibility

- Reproduce: `pip install -r requirements-round2.txt && ./run_round2.sh`
- Pipeline: `src/round2/{data,train,evaluate,figures,build_eval_report,build_tech_report,predict}.py -- one seed (42) throughout.`
- Artefacts: `models/round2/*.pkl`, `output/round2/metrics.json` + figures + PDFs, `notebooks/02_nlp_rebuild_semantic_layer.ipynb` (executed).
- Inference demo: `python src/round2/predict.py --text "..."`

11 · Conclusion

The semantic layer is rebuilt: `logreg_union` for sentiment (test macro-F1 0.6134) and `svc_union` for topic (test macro-F1 0.8051), selected by CV on train, confirmed once on held-out test, with calibrated probabilities, measured failure slices and published mistakes. Every claim above traces to `output/round2/metrics.json`.